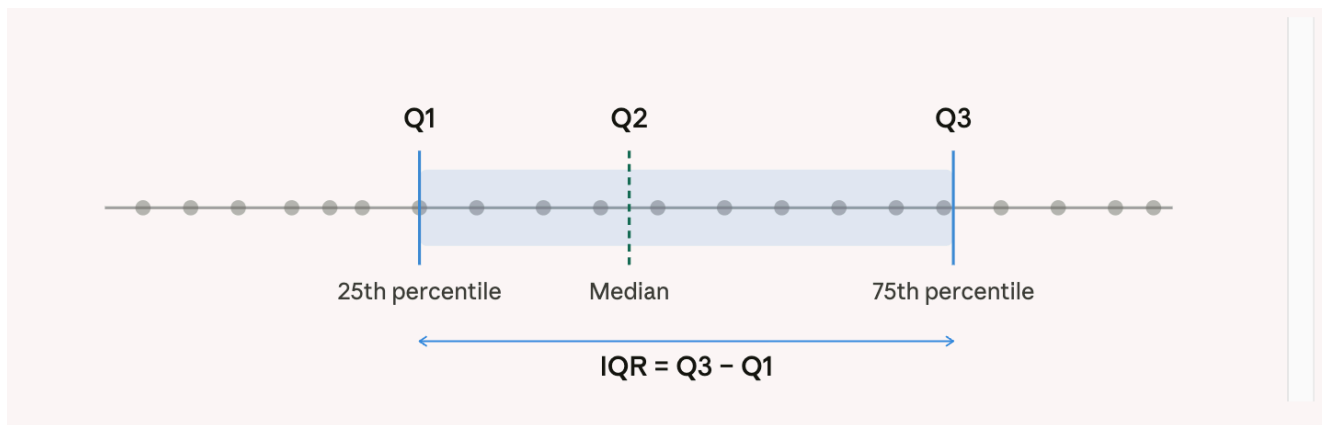


IQR-Based Capping

When we encounter the outlier, it is a pain in the neck

instead of removing the outlier (which loses data), capping clips them to a reasonable boundary. The IQR method gives you a principled, data-driven way to define those boundaries.

Interquartile Range (IQR) measures the spread of the middle 50% of the data



Q_1 = the value below which 25% of the data falls

Q_3 = the value below which 75% of the data falls

$IQR = Q_3 - Q_1$ = the range of the middle half of your data

Calculate the fence/bound

Using the IQR, you compute two **boundary fences**. Any value beyond these fences is considered an outlier:

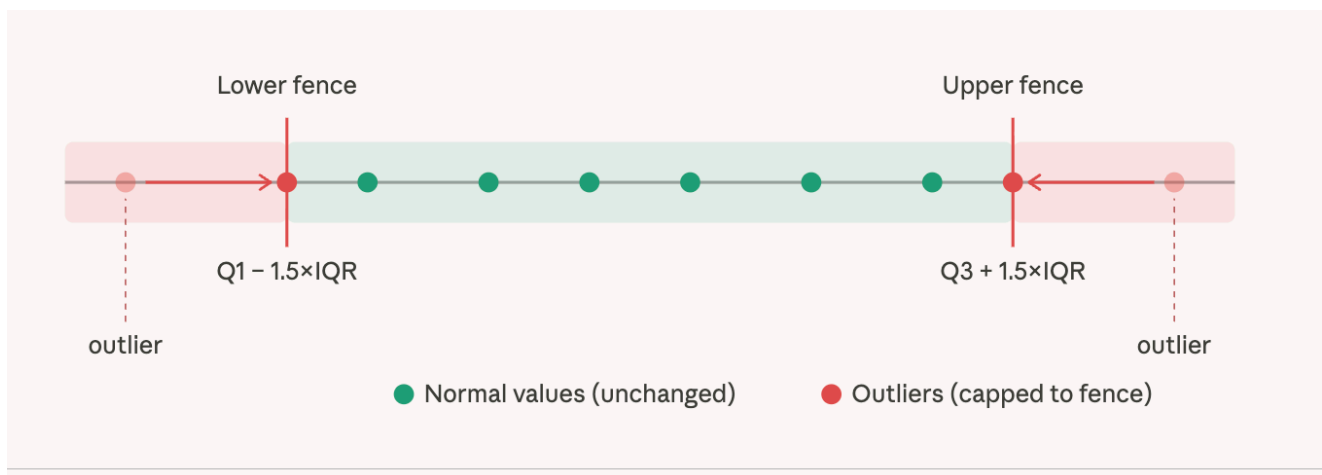
$$Lowerbound = Q1 - 1.5 \times IQR$$

$$Upperbound = Q3 + 1.5 \times IQR$$

The multiplier **1.5** is the standard choice (originally proposed by John Tukey for box plots). Some use **3.0** for more extreme outliers only.

Capping = clipping/Winsorizing cap

Then we apply the cap, instead of removing the outlier, we replace them with the fence value



- Any value **below** the lower fence → replaced with the lower fence
- Any value **above** the upper fence → replaced with the upper fence
- Everything in between → left unchanged

```
def cap_outliers(data, column):

    Q1 = data[column].quantile(0.25)

    Q3 = data[column].quantile(0.75)

    IQR = Q3 - Q1

    lower_bound = Q1 - 1.5 * IQR

    upper_bound = Q3 + 1.5 * IQR

    data[column] = data[column].clip(lower_bound, upper_bound)

    return data

for col in ['Tumor Size', 'Regional Node Examined', 'Regional Node Positive']:

    cap_outliers(df_numeric, col)
```